

```
from google.colab import files
```

```
uploaded = files.upload()
```

[Choose Files](#) Telco-Custo...r-Churn.csv

Telco-Customer-Churn.csv(text/csv) - 977501 bytes, last modified: 9/27/2019 - 100% done
Saving Telco-Customer-Churn.csv to Telco-Customer-Churn.csv

```
import pandas as pd
```

```
df = pd.read_csv('Telco-Customer-Churn.csv')
```

```
df.head()
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	
1	5575-GNVDE	Male	0	No	No	34	Yes	No	
2	3668-QPYBK	Male	0	No	No	2	Yes	No	
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	
4	9237-HQITU	Female	0	No	No	2	Yes	No	

5 rows × 21 columns

```
print(df.columns)
```

```
df.info()
```

```
Index(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',
      'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
      'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',
      'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling',
      'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'],
      dtype='object')
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerID            7043 non-null  object
1   gender                 7043 non-null  object
2   SeniorCitizen          7043 non-null  int64
3   Partner                7043 non-null  object
4   Dependents             7043 non-null  object
5   tenure                 7043 non-null  int64
6   PhoneService           7043 non-null  object
7   MultipleLines          7043 non-null  object
```

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8  InternetService  7043 non-null  object
9  OnlineSecurity  7043 non-null  object
10 OnlineBackup    7043 non-null  object
11 DeviceProtection 7043 non-null  object
12 TechSupport     7043 non-null  object
13 StreamingTV     7043 non-null  object
14 StreamingMovies  7043 non-null  object
15 Contract        7043 non-null  object
16 PaperlessBilling 7043 non-null  object
17 PaymentMethod   7043 non-null  object
18 MonthlyCharges  7043 non-null  float64
19 TotalCharges    7043 non-null  object
20 Churn           7043 non-null  object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB

```

```

df = df.drop('customerID', axis=1)

df['Churn'] = df['Churn'].map({'Yes':1, 'No':0})

df = pd.get_dummies(df, drop_first=True)

df.head()

```

	SeniorCitizen	tenure	MonthlyCharges	Churn	gender_Male	Partner_Yes	Dependents_Yes	PhoneSe
0	0	1	29.85	0	False	True	False	
1	0	34	56.95	0	True	False	False	
2	0	2	53.85	1	True	False	False	
3	0	45	42.30	0	True	False	False	
4	0	2	70.70	1	False	False	False	

5 rows × 6560 columns

```
df.isnull().sum()
```

	0
SeniorCitizen	0
tenure	0
MonthlyCharges	0

```
from sklearn.model_selection import train_test_split

X = df.drop('Churn', axis=1)
y = df['Churn']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

print("Training set:", X_train.shape)
print("Test set:", X_test.shape)
```

Training set: (5634, 6559)
 Test set: (1409, 6559)

TotalCharges 999.9 0

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

model = RandomForestClassifier(random_state=42)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

Accuracy: 0.7977288857345636

Confusion Matrix:
 [[957 79]
 [206 167]]

Classification Report:

	precision	recall	f1-score	support
0	0.82	0.92	0.87	1036
1	0.68	0.45	0.54	373
accuracy			0.80	1409
macro avg	0.75	0.69	0.70	1409
weighted avg	0.78	0.80	0.78	1409

